SECTION 536
PAVEMENT CRACK SEALING

DESCRIPTION

536.01 Scope – This work consists of restoring the asphalt concrete pavement surface to a less permeable condition. The work includes routing when requested, cleaning and filling the crack with sealant, and dusting or sanding.

536.02 Purpose – The purpose of crack sealing shall be to prevent moisture from penetrating into the road base through surface cracks and to extend pavement life. This shall be carried out by effectively sealing the cracked surface with a highly rubberized and elasticized or high float asphalt sealant product.

MATERIALS

536.11 Asphaltic – The rubberized asphaltic and/or elasticized asphalt sealant products shall meet the requirements listed in Table 536-A.

It shall be noted that ASTM D6690 requirements are not mandatory in the Lower Mainland, Vancouver Island or the Sunshine Coast.

As specified in the Special Provisions the Contractor shall use a Rubberized or High Float Emulsified Asphalt HF150S or, with the approval of the Ministry Representative, either HF 150P (hot pour) or HF 250S (cold pour).

Table 536-A: Sealant Requirement

ASTM Test	Description
D6690	Specification for joint and crack sealants, hot applied, for concrete and asphalt pavements

536.12 Sand – On pavements that require the use of High Float Emulsified Asphalt, the Contractor shall supply "blinding sand." The sand, when tested according to ASTM C117, shall meet the gradation requirements shown in Table 536-B and shall be approved by the Ministry Representative.

536.13 Dust Cover – Pavements that are sealed with rubberized or elasticized sealant shall receive a dust coating with a material such as Portland Cement, talc, lime or the equivalent material approved by the Ministry Representative.

Table 536-B: Blinding Sand Gradation Requirements

Sieve Size (mm)	% Passing
6.3	100
4.75	95 - 100
2.36	50 - 100
0.600	20 - 70
0.300	5 - 25
0.075	0 - 3

EQUIPMENT

536.21 Router – Where the cracks require routing, the Contractor shall provide a vertical router capable of routing asphaltic pavements to a depth of 25 mm and a width of 16 mm. It shall be capable of a minimum production of 200 metres per hour, even when following meandering cracks without unnecessary pavement cutting.

536.22 Cleaner – The cleaner unit shall be a hot compressed air fed propane fired device capable of cleaning, heating and drying routed cracks with not less than 1.7 m^3/min of compressed air at 690 kPa. It shall have valving/hoses and fittings for the mixture of liquid propane gas and compressed air. The hot air exhaust shall not exceed 315°C.

536.23 Melter – The melting kettle shall be of the double boiler type supplying indirect heating to slowly heat the asphaltic material with heat transfer oil. It shall have built in calibrated thermometers for both transfer oil and the sealing compound. The heating of the sealant shall be efficient and thermostatically controlled so as not to exceed the manufacturer's maximum safe heating temperature and shall be such as to maintain a constant temperature once the sealant is heated. The melter shall be capable of constantly agitating the asphaltic material as it is being heated and shall have a pump circulating the sealant from the bottom to the top of the kettle.

The melter must comply with the Gas Safety Act Regulations and Codes and any other applicable acts and regulations.

536.24 Filler Tools – Crack filler devices and strike off tools must be such that successful forming of the bead of sealant over the prepared crack is as specified in this Section.

CONSTRUCTION

536.31 General – Crack sealing shall only be performed when the pavement surfaces are dry, and the crack and road base are dry or nearly dry (no visible moisture), and the surface temperature shall be 10°C or higher.

The Ministry Representative shall select a random sample of virgin sealant from each Lot for testing purposes.

Where required by the Contract, cracks up to 16 mm in width shall be widened by using a router to form a sealant reservoir 16 mm in width and from 19 mm to 25 mm in depth. All routing shall be performed keeping the crack centreline within ± 8 mm of the centre of the rout and shall be cleaned with a hot compressed air lance. Material removed from the cracks shall be disposed of in an environmentally compliant and sound manner as approved by the Ministry Representative.

The routing speed shall be such that the pavement is carefully cut, not broken or torn out, and the sides of the rout are smooth and uniform. The surface of the pavement and routed crack shall be cleaned of all dust and routing debris.

Routing should not be carried out on pavements that are of such an age that pavement fractures or spalls occur along the edge of the freshly routed crack. Normally pavement fractures or spalls should not occur unless pavements are in excess of ten (10) years old. The decision to forgo routing shall be made by the Ministry Representative.

The crack shall be filled with sealant from the bottom to the surface level in such a manner that the sealant does not bridge entrapped air pockets. Material shall be placed to overfill the crack. It will then be struck off to leave a uniform amount of sealant directly over the crack, with the edges of the spread evenly feathered to overlap on the pavement surface from a minimum of 25 mm to a maximum of 40 mm on each side of the crack. The sealant overband shall not be so thick that it can be removed during snow plowing or produce a noticeable bump when traversed by traffic.

The Contractor shall repair any damage done to the sealant by stones or any other deleterious material being embedded in the sealing compound.

The Contractor shall ensure that traffic is kept off the sealed cracks until the sealant has properly set up and will not be damaged or pulled out by the passage of traffic.

536.32 Rubberized and Elasticized Asphalt Sealants – These shall be used on pavements that are less than seven (7) years old and where the majority of cracks are less than 25 mm in width.

The sealant shall be applied to cracks that have been routed to a uniform depth and width.

Cracks having a width greater than 16 mm need not be routed but shall be cleaned to a minimum depth of 25 mm.

Within two minutes of the completion of the cleaning operation, the crack shall be filled with sealant from a melter using a connecting wand or manual applicator (pouring cone) which ensures minimum pour application temperatures for the product are maintained.

Upon completion of the pouring, the sealant shall be dusted to prevent the asphalt from tracking. Excess dusting material shall be removed.

536.33 High Float Emulsified Asphalt – High Float Emulsified Asphalt shall be used on pavement where there are depressions or lipping at the cracks, or majority of cracks are 5 mm to 35 mm in width.

The cracks shall be cleaned as close to the actual depth as possible and the removed material shall be disposed of in an environmentally compliant and sound manner as approved by the Ministry Representative. Immediately after cleaning, they shall be filled with sealant from a distributor truck or melter. The distributor truck or melter shall have an efficient means of heating the sealant to any temperature up to 100°C and maintaining it constantly at the manufacturer's prescribed temperature without overheating.

Upon completion of the sealing of the crack, the sealant shall be sanded to prevent the asphalt from tracking. Excess sand shall be removed and disposed of, by the Contractor, in an environmentally compliant and sound manner as approved by the Ministry Representative.

536.34 Warranty – The Contractor shall rectify any defects identified during the Warranty Period within twenty-one (21) days from the time first detected by or reported to the Contractor to the satisfaction of the Ministry Representative.

MEASUREMENT

536.81 General – Pavement Crack Sealing will be measured by the LITRE using weight slips.

PAYMENT

536.91 General – Payment for PAVEMENT CRACK SEALING will be at the Contract Unit Price bid per litre. The Unit Price shall be full compensation for all labour, equipment and materials supplied as required to rout or clean with compressed air and seal the pavement cracks, as specified.

SECTION 537

PAVEMENT SPRAY PATCHING

DESCRIPTION

537.01 Scope – This specification describes the materials and the professional standards required for supplying and applying tack coat and a mixture of asphalt emulsion and aggregate or a pre-mixed 'spray patch' material for repairing pavement cracks and other defective locations on the pavement surface as applicable.

537.02 Purpose – The purpose of spray patching is to repair potholes, depressed pavements and cracks wider than 35 mm.

MATERIALS

537.10 537.10 Asphalt Materials – The asphalt emulsion products shall meet the requirements listed in Table 537-A. the Contractor shall ensure that the asphalt emulsion is compatible with the aggregates used for spray patching.

537.11 Aggregate – The Contractor shall supply the aggregate necessary to produce a dense, stable and durable patch. The aggregates shall meet the requirements specified herein.

537.11.01 Test Methods – The aggregate will be sampled and tested in accordance with the standard ASTM procedures and tests listed in Table 537-B.

537.11.02 Test Requirements – The aggregate shall be clean, hard, durable, moderately sharp, and free from coatings of clay, silt or other deleterious material and shall contain no clay balls or other aggregations of fine material. The aggregates shall not be markedly hydrophilic in character.

The aggregate, when tested by the methods listed in Table 537-B, shall meet the following criteria:

(a) Clay Lumps – The aggregate shall not contain more than 0.5% by mass of clay lumps or other aggregation of fine material.

(b) Micro-Deval – No individual test shall have a loss factor of more than 20%.

(c) Plasticity – All aggregate shall be non-plastic when tested in accordance with ASTM D4318.

537.12 537.12 Aggregate Gradation – The aggregates shall meet the gradation requirements of Table 537-C.

537.13 Spray Patching Mix – The spray patching mix used shall exhibit uniform coating, good cohesion, stability and workability.

Table 537-A:Emulsion Requirement for Spray Patching

AASHTO Test	Emulsion Property Limits	
	RS-2	WSPE[1]
T 49: Penetration, 100 g, 5 s	100 - 200	90 - 150
T 50: Float test, 60°C, seconds	–	≥ 1200
T 59:		
Viscosity, Saybolt Furol, 50°C, seconds	75-400	≥ 50
Demulsibility, 0.02 N $CaCl_2$, %	60 min	≤ 65 max
Sieve Test, %.	≤ 0.1	≤ 0.1
Residue by Distillation, %	≥ 65 min	≥ 68
Oil Distillate, Volume %	–	≤ 7

Note 1: Winter Grade Spray Patch Emulsion

Table 537-B: Test Methods for Mineral Aggregate

ASTM Designation	Name of Test
D75	Standard Practice for Sampling Aggregates
C117	Standard Test Method for Materials Finer than 75-µm (No. 200) Sieve in Mineral Aggregates by Washing
C136	Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
C142	Standard Test Method for Clay Lumps and Friable Particles in Aggregates
D4318	Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
D6928	Standard Test Method for Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus

Table 537-C: Gradation Requirements for Spray Patching

Sieve Size (mm)	% Passing
9.5	100
4.75	75-100
2.36	0-30
1.18	0-10
0.075	0-5

EQUIPMENT

537.21 Patching Unit – The Contractor shall supply all equipment necessary to complete the Work including a patching unit of a design currently accepted by the Ministry and in use for highway maintenance.

The machine shall:

(a) provide a minimum of 5.0 m^3/min of compressed air to clean cracks and the surface to be patched; and

(b) be capable of:

(i) spraying the asphalt emulsion to provide a tack coat;

(ii) blending aggregate and emulsion;

(iii) applying the resulting mix to provide a high-density patch; and

(iv) covering the repaired area with dry aggregate

The Contractor shall also supply appropriate compaction equipment, suitable for the mix, that is acceptable to the Ministry Representative.

CONSTRUCTION

537.31 General – Spray patching shall only be performed when the pavement surfaces are dry, and the crack and road base are dry or nearly dry (no visible moisture), and the pavement surface temperature shall not be lower than 10°C.

The cracks and surface on which the spray patching material to be placed shall be free of dirt, sand, foreign matter, and loose material, and be lightly tack coated with emulsion.

All cracks, potholes and depressions shall be filled to the level of surrounding surface, producing a level, rideable patch, free of excess asphalt and, when swept, free of loosening rock.

The repaired area shall be compacted to ensure adequate embedment of the asphalt aggregate mixture into and over the crack or depression.

Where the in-place mix exhibits ravelling or excessive tracking, the patch shall be covered with dry aggregate.

QUALITY CONTROL AND QUALITY ASSURANCE

537.41 Quality Control by the Contractor – The Contractor shall be responsible for providing all resources required to carry out the quality control on all the constituent materials, processes and products, their testing and inspection, within the construction of the spray patch and the quality of the end product.

537.42 Sampling and Testing – The Ministry may request the Contractor to provide test results showing that the materials used for spray patching meet the requirements of this specification as determined by the applicable tests.

The Contractor shall be responsible to provide representative samples for the aggregate, emulsion and patching mixture as required when requested by the Ministry. These samples shall be taken in presence of the Ministry Representative.

The above samples may be tested by the Ministry to ensure that they meet the requirements of this specification.

537.43 Acceptance – Spray patches shall be accepted based on following.

(a) visual observation of the patching material,

(b) surface appearance of the patch at the time of patching,

(c) continuity with adjacent surfaces, and

(d) conformance of the patching materials to the requirements of this specification.

At the request of the Contractor, the Ministry Representative will attend on-Site to observe the initial spray patching and advise on its acceptability, before the Contractor proceeds with the balance of the Work.

Prior to issuance of the Completion Certificate, all finished patches shall be free of defective areas, as determined by the Ministry Representative, including but not limited to, wheel track marking, ravelling, uneven surface, and fat spots.

Areas showing defects shall be deemed Unacceptable Work and shall be repaired by the Contractor at no cost to the Ministry.

MEASUREMENT

537.81 General – Pavement Spray Patching will be measured by the volume of emulsion actually sprayed in LITRES.

PAYMENT

537.91 General – Payment for spray patching shall be made at the Contract Unit Price per LITRE.

The supply and application of aggregate is incidental to the Work and will not be paid separately.

Payment at the Contract price for the above items shall be full compensation for all labour, equipment, and material to do the Work.

SECTION 541
STONE PAVING

DESCRIPTION

541.01 Scope – This Section covers the construction of stone paving for dry-laid and grouted slope protection, culvert and spillway aprons, ditches and other similar uses.

A minimum thickness of 100 to 225 mm will be specified for paving rock depending upon location, use and setting requirements. Requirements for rock exceeding 200 mm thickness are specified by SS 205 for hand-laid and grouted riprap.

MATERIALS

541.11 Materials

541.11.01 Rock – Rock for stone paving shall generally be obtained from the roadway excavation, stockpiled where designated and selected as necessary for use at the locations indicated on the Drawings, unless the supply of off-site rock is specified.

Rock will be required as either:

(a) Angular rock as found or blasted; or

(b) Rounded rock.

All rock shall be sound, durable and clean, relatively flat on the face to be exposed, reasonably consistent in colour tone, of acceptable size (200 mm to 450 mm across) and of the required thickness range.

541.11.02 Gravel Base – Gravel base shall be 25 mm well-graded crushed granular aggregate.

541.11.03 Bedding Mortar – Bedding mortar shall consist of one part Portland cement to three parts clean, well-graded fine aggregate or coarse sand mixed with water to give a 150 mm slump.

541.11.04 Setting and Jointing Mortar – Setting and jointing mortar shall consist of one part Portland cement to three parts clean, well-graded fine aggregate mixed with water to give a 75 mm slump.

CONSTRUCTION

541.31 Preparation – Graded surfaces to receive paving shall be trimmed and well tamped to the lines and grades indicated on the Drawings and to the approval of the Ministry Representative.

Edges of paving shall be constructed of thicker stones to form a stable foundation and protection against any undercutting in waterways.

All paved areas shall receive a minimum 150 mm layer of gravel base, well consolidated.

Gravel level inside curbs shall permit the surface of the jointing mortar to be flush with the top of the curb.

541.32 Construction – Construction shall be carried out with all material, labour, tools, equipment and incidentals supplied by the Contractor as necessary to complete all stone paving work in accordance with good work practice.

A gravel base of grouted stone paving shall be well saturated with bedding mortar slurry.

All rock shall be firmly embedded into the base material so that the voids between stones are of minimal width with the rock surfaces relatively even as indicated on SS Drawing SP541-01.

Voids of grouted paving rock shall be filled with setting and jointing mortar, 75 mm thick and, after consolidation and any necessary chinking of the larger voids, the mortar surface shall be smooth finished and sloped to drain freely.

Finished stone paving shall show a clean even appearance, free of disfiguring spalls and, where grouted, cleaned of mortar staining.

MEASUREMENT

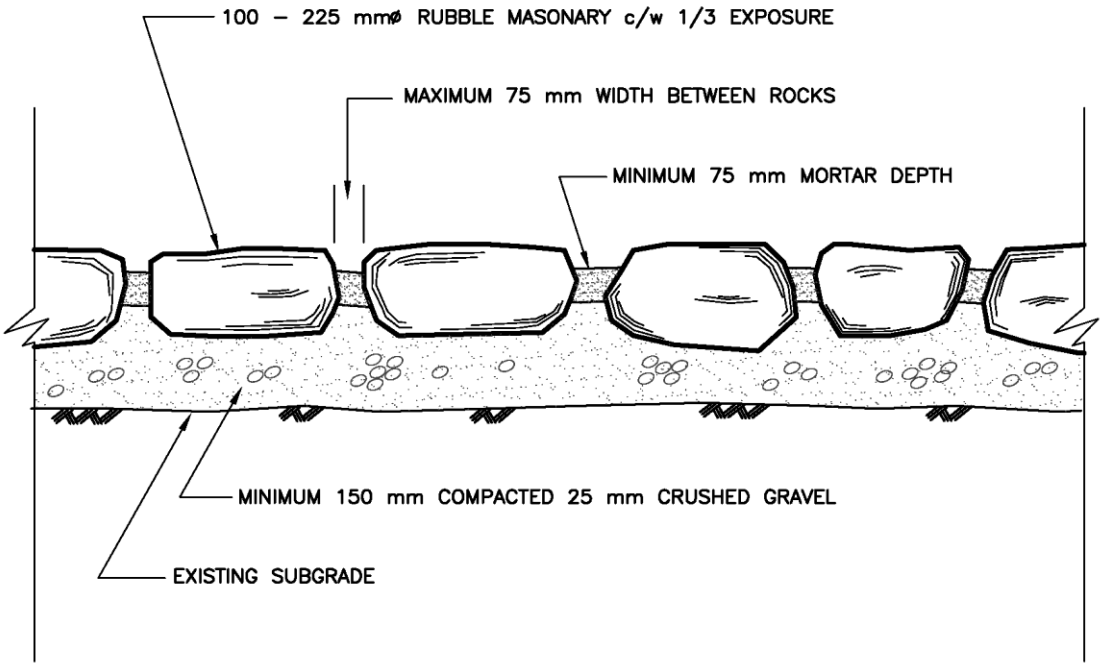
541.81 General – Stone Paving work will be measured by the SQUARE METRE of stone in place.

PAYMENT

541.91 General – Payment for STONE PAVING work will be at the Contract Unit Price per square metre of stone in place.

The Contract Unit Price shall be accepted as full compensation for the supply of the gravel base, stone and mortar, for complete installation, including trimming and incidental excavation, compaction, finishing, and for all subsidiary work not required to be separately paid for.

TYPICAL SECTION THRU ROCK PAVING	SP541–01



NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 582
CONCRETE CURB AND GUTTER AND STORM DRAINAGE

DESCRIPTION

582.01 Scope – This Section describes the requirements for construction of roadside curb and gutter and associated underground drainage systems. Such work shall consist of constructing and/or installing appurtenances shown or indicated on the drawings and Schedule of Approximate Quantities and Unit Prices.

582.02 Referenced Specifications – Unless otherwise specified, the most recent published metric edition applies.

AASHTO:

- M 306 Standard Specification for Drainage, Sewer, Utility, and Related Casting
- LRFD Bridge Design Specifications
- Standard Specifications for Highway Bridges

ASTM:

- A48 Standard Specification for Gray Iron Castings
- A536 Standard Specification for Ductile Iron Castings
- C14 Standard Specification for Nonreinforced Concrete Sewer, Storm Drain, and Culvert Pipe
- C76 Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe
- C309 Standard Specification for Liquid Membrane Forming Compounds for Curing Concrete
- C443 Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
- C478 Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
- C506 Standard Specification for Reinforced Concrete Arch Culvert, Storm Drain and Sewer Pipe
- C507 Standard Specification for Reinforced Concrete Elliptical Culvert, Storm Drain, and Sewer Pipe
- C1218 Standard Test Method for Water-Soluble Chloride in Mortar and Concrete
- C1433 Standard Specification for Precast Reinforced Concrete Monolithic Box Sections for Culverts, Storm Drains, and Sewers
- C1677 Standard Specification for Joints for Concrete Box, Using Rubber Gaskets

CSA:

- A23.1 Concrete materials and methods of concrete construction
- A23.2 Test Methods and standard practices for concrete
- A23.4 Precast Concrete – Materials and Construction
- A257 Standards for concrete pipe and manhole sections
- A3000 Cementitious Materials Compendium
- S6 Canadian Highway Bridge Design Code (CHBDC)

MATERIALS

582.10 Design of Underground Drainage System

582.10.01 Products – Underground drainage system products shall be designed in accordance with the applicable design codes, standards and referenced specifications, and the following:

(a) Highway Live Load – Underground drainage system products shall be designed for highway design live load as follows unless otherwise specified herein:

(i) The design live load in accordance with the Ministry Supplement to the CHBDC S6, or

(ii) AASHTO LRFD Bridge Design Specifications HL-93 live load, or

(iii) AASHTO Standard Specifications for Highway Bridges H-20 or HS-20 live load

(b) Direct Contact Loading – The highway design live load for components that may be subjected to direct contact with traffic wheels shall also include dynamic impact effects. The dynamic impact allowances shall be in accordance with the applicable design codes and standards.

582.11 Drain Pipe – All drain pipe shall be of first quality, sound, true in form and free from defects of all kinds.

582.12 Manufacture of Precast Concrete Storm Drainage Products – All concrete pipe, box culverts, manholes, catch basins, precast concrete endwalls and

associated precast concrete products shall be supplied from a manufacturer that is certified by an independent third party certification agency to produce these products according to applicable CSA and ASTM manufacturing standards.

582.12.01 Certified Manufacturer – The manufacturer shall be certified under at least one for the following quality assurance programs:

(a) The Canadian Precast Concrete Quality Assurance (CPCQA) Certification Program.

(b) The CSA Group Testing & Certification Inc. (CSA) Certification Program

(c) QCAST Certification Program by the American Concrete Pipe Association for Precast Products.

(d) National Precast Concrete Association (NPCA) Plant Certification.

(e) Other certification organizations acceptable to the Ministry Representative that are accredited by the Standards Council of Canada for the certification of manufacturers for these products.

582.12.02 Product Markings – All products shall be marked with the certification body's logo confirming that the production of the product is in accordance with the quality and requirements of the specified standards for the products.

582.13 Concrete Pipe and Boxes

582.13.01 Reinforced Concrete Pipe – Reinforced concrete pipe shall be designed for flexible rubber gaskets and shall meet the requirements of CSA A257, ASTM C76, ASTM C506, ASTM C507, or ASTM C655.

582.13.02 Plain Concrete Pipe – Plain concrete pipe shall be designed for flexible rubber gaskets and shall meet the requirements of CSA A257 or ASTM C14.

582.13.03 Concrete Box Culverts – Concrete box culverts shall conform to the requirements of ASTM C1433 or ASTM C1577.

582.13.04 Gaskets – Gaskets shall be supplied with all concrete pipe and boxes and shall meet the requirements of CSA A257, ASTM C443, or ASTM C1677, all as appropriate to the application.

582.13.05 Requirements – Pipe and boxes shall be legibly marked with the date of manufacture, the name or trademark of the manufacturer and by marks denoting plain concrete pipe, reinforced concrete pipe or extra strength reinforced concrete pipe, as the case may be. Pipe containing elliptical reinforcement or otherwise requiring special placement shall be marked on the inside of the pipe with the words "Top" or "Bottom" at the correct place to indicate the proper position when installed.

Upon request, the Contractor shall furnish at the Contractor's expense such test and other information as may be required regarding the concrete pipe proposed to be used.

582.14 Corrugated Steel Pipe – Corrugated steel pipe shall conform to the requirements of SS320.

582.15 Catch Basins and Manholes

582.15.01 Precast Reinforced Concrete – Precast reinforced concrete catch basin and manhole products, including but not limited to, barrels (risers), grade rings, flat slab tops (lids), conical sections, base sections, steps and ladders shall meet the requirements of ASTM C478 or CSA A257, and shall be gasketed or grouted to prevent infiltration and exfiltration.

582.15.02 Corrugated Metal – Corrugated metal barrels shall be in accordance with SS 320.

582.15.03 Covers and Frames – Catch basin and manhole covers and frames shall be cast iron and meet the requirements of ASTM A48. Catch basin and manhole covers and frames shall be designed for highway loading in accordance with either of the following:

(a) AASHTO M 306 using the H-20 or HS-20 proof load testing with the proof load of 178 kN; or

(b) The design live load in accordance with the Ministry Supplement to the CHBDC S6, including the dynamic impact allowance, with the wheel load application distributed over an area of 250 mm x 250 mm.

582.16 Detectable Warning Mats

582.16.01 General – Detectable warnings shall consist of a surface of truncated domes, sized and spaced as specified below and as shown in Figure 582-1.

582.16.02 Dome Size – Truncated domes in a detectable warning mat surface shall have a base diameter of 2 mm minimum and 36 mm maximum, a top diameter of 50 percent (50%) of the base diameter minimum to 65 percent (65%) of the base diameter maximum, and a height of 5.1 mm.

582.16.03 Dome Spacing – Truncated domes in a detectable warning mat surface shall have a center-to-center spacing of 41 mm minimum and 61 mm maximum, and a base-to-base spacing of 17 mm minimum, measured between the most adjacent domes on a square grid.

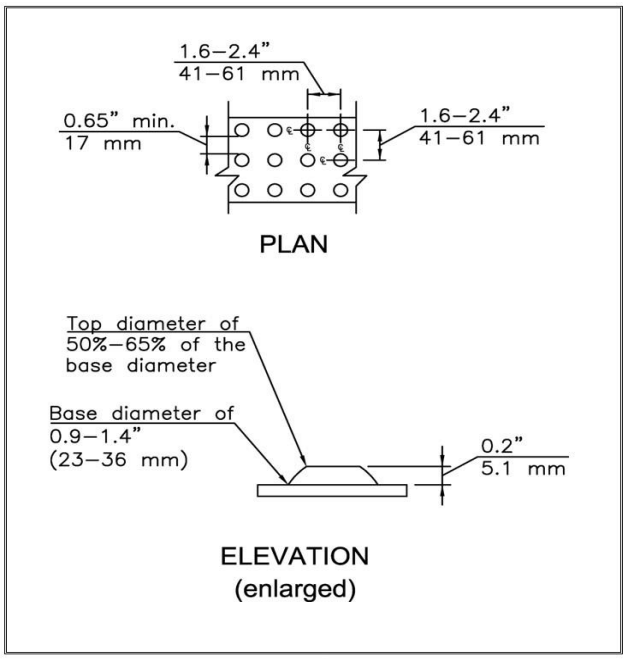
582.16.04 Contrast – Detectable warning mat surfaces shall contrast visually with adjacent walking surfaces either light-on-dark, or dark-on-light.

582.17 Cast-in-Place Concrete – All cast-in-place [CIP] concrete shall be in accordance with SS 211 except as modified herein.

582.17.01 CIP Concrete Materials – Concrete material properties shall be as shown on the SS 582 SP Drawings.

Contrary to SS 211.04.05, aggregate tests CSA A23.2-14A and CSA A23.2-25A are not required.

Figure 582-1:Detectable Warning Mats – Dome Size and Spacing



582.17.02 Concrete Mixes

(a) (a) CIP Concrete Mix Specifications – All cast-in-place concrete used under this SS 582 shall have the properties specified in Table 582-A.

(b) Mix Design – The Contractor shall submit the mix design documentation in accordance with SS 218, Concrete for Minor Works.

(c) Trial Mixes – Contrary to SS 211.03.05, trial mixes are not required.

Table 582-A:Cast-in-Place Concrete Mix Properties

Criteria	Placement Method	
	Hand	Extruded
Minimum 28-Day Compressive Strength	32 MPa	
Maximum w/cm ratio by mass	0.45	
Nominal Maximum Size of Coarse Aggregate	20 mm	10 mm
Entrained Air	5 to 8%	6 to 9%
Maximum Slump	80 mm	25 mm

582.17.03 Sulphate Exposure – The Ministry shall notify the Contractor if S-1 and/or S-2 exposure conditions exist for the soils present on-Site. If no information is provided by the Ministry, S-3 shall be assumed.

For imported bedding material, the aggregate supplier shall perform the sulphate and chloride exposure assessment of its aggregate in accordance with testing requirements provided in Table 582-B.

The test results will be used to determine the site-specific exposure level as defined in Table 3 of CSA A23.1. Cementitious materials to be used shall be in accordance with the Table 3 of CSA A23.1, and be tested using CSA A3004-C6 for portland / portland-limestone cements or A3004-C8 for blended hydraulic cements or MS / HS equivalent cements to meet the requirement for the exposure level in the table.

Table 582-B: Testing Requirements for Sulphate Exposure

Bedding Material	Test Parameter	Applicable Test Methods
Recycled Concrete Aggregate	Sulphate content	CSA A23.2-8B
Other Aggregates	Sulphate content	CSA A23.2-3B
	Chloride Content	ASTM C1218*

* Note: Typically used for mortar or concrete but can be adapted for soil.

582.18 Self-Consolidating Concrete – Self-Consolidating Concrete (SCC) may be used in the manufacture of precast concrete pipe (reinforced or plain), box culverts, manhole and catch basin barrels, grade rings, flat slab tops, risers, conical sections, and base sections only under the following conditions:

582.18.01 SCC Limitations – SCC may only be used for products limited to the following sizes and specifications

(a) pipes and barrels smaller than 3.0 m in any internal cross-sectional dimension manufactured in accordance with CSA A257, ASTM C14, ASTM C76, ASTM C478, ASTM C506, ASTM C507, or ASTM C655,

(b) boxes with both internal spans and rise less than 3.0 m manufactured in accordance with ASTM C1433 or C1577,

(c) with written approval from the Ministry Representative, culvert endwalls manufactured in accordance with CSA A23.4

(i) less than 2.0 m in exposed height; or

(ii) where Ministry unique structure numbers are precast in, up to 3.5 m of exposed height.

582.18.02 SCC Documentation – The Contractor shall provide the following for review upon request by the Ministry Representative:

(a) a full copy of the applicable concrete mix design utilized in the manufacture of any product(s) intended to be incorporated into Ministry works; and

(b) Documentation in accordance with either of the following:

(i) Details in accordance with CSA A23.2-24C documenting the proven history of successful performance for the required strength, durability, and other performance requirements of the proposed mix design for the manufacture of any product(s) intended to be incorporated into Ministry works, or

(ii) A full-scale test shall be used to verify the self-consolidating characteristics for placement and for the hardened concrete properties of the mix design for the manufacture of any product(s) intended to be incorporated into Ministry works. Documentation in accordance with CSA A23.2-24C shall be submitted demonstrating that the proposed mix design will achieve the required strength, durability, and performance requirements

(c) The self-consolidating concrete mix shall conform to the requirements of CSA A23.1, including Clause 8.6 and Table 22, and the appropriate CSA and ASTM standards listed in SS 582.18.01 above.

CONSTRUCTION

582.31 Concrete Construction – All concrete work performed under this SS 582 shall meet the requirements of SS 211 except as modified herein and be constructed to the line and grade shown on the Drawings or as otherwise directed by the Ministry Representative. Combined concrete curb and gutter will include returns at the intersections if so directed.

The combined curb and gutter shall be constructed to the standard cross-section shown on SS Drawing SP582-01.01 and to such layout plans as are attached, except where the face of the curb varies and at lanes and driveways where drop curbs are ordered.

Expansion joints 13 mm in thickness cut to the cross-section of the concrete curb and sidewalk, composed of premoulded joint filler shall be installed at approximately 15 m intervals for sidewalks and 9 m intervals for concrete curb. They will also be installed when the sidewalk abuts a rigid structure or changes direction as well as at curb returns, either side of driveways, and other locations as directed by the Ministry Representative. Expansion and isolation joints shall be constructed to the full depth of the slab. Contraction joints shall be placed to a maximum of 3 m intervals, across both sidewalk and concrete curb and be tooled or saw-cut into the concrete to a depth of 1/3 the thickness of the concrete.

Concrete Curb horizontal and vertical alignment shall be in accordance with the Drawings within ±10 mm, and such that there is no gap greater than 6 mm under a 3 m straight edge.

Ambient temperature forecasts used to trigger hot weather or cold weather concreting procedures shall be as provided by Environment Canada or another source acceptable to the Ministry Representative.

582.31.01 Delivery – Contrary to SS 211.08.03, the temperature of all concrete, from time of batching to complete discharge shall remain between 10°C and 32°C.

582.31.02 Finishing – All cast-in-place concrete shall be finished with a magnesium or wood float, to a closed, uniform surface. Sidewalks shall be textured transversely with a medium broom finish.

582.31.03 Curing – When ambient temperatures are, or are forecast to be, greater than 35°C at any time during the seven (7) days after placement, curing shall be done in accordance with SS 211.15.02.

At temperatures below 35°C, the Contractor shall provide concrete curing using:

(a) Moist curing per SS 211.15.02; or

(b) Poly-Alpha methyl-styrene (PAMS), Type 2, Class B, or Acrylic Cure and Seal, ASTM C309 based curing compound to all exposed concrete surfaces at rate recommended by manufacturer in two (2) applications within 12 hours of placing concrete.

Resin- and wax-based curing compounds shall not be used.

582.31.04 Hot Weather Concreting – Hot Weather Concreting procedures shall be done in accordance with SS 211.18 (except as modified herein) when ambient air temperature is, or is forecast to be, 35°C or higher during the placing period.

Concrete shall not have a temperature greater than 25°C at any time between batching and placement. Concrete shall not be placed against any material/surface which is at a temperature above 35°C.

Curing shall be in accordance with SS 582.31.03 and commence as soon as possible.

582.31.05 Cold Weather Concreting – Cold Weather Concreting procedures shall be done in accordance with SS 211.19 (except as modified herein) when ambient air temperature is, or is forecast to be, below 5°C during placement, and/or is forecast to fall below 5°C during the first seven (7) days after placing.

Concrete shall not be placed against any material/surfaces which is at a temperature below 0°C.

Contrary to SS 211.19(a), the temperature of the concrete shall be between 15°C and 30°C, from the time of batching until placement.

Contrary to SS 211.19(e), the Contractor shall provide protection and heating if required, to:

(a) maintain a uniform curing temperature, at a minimum 10°C, for at least three (3) days; and

(b) protect the concrete from freezing for another four (4) days.

582.31.06 Repairs – Contrary to SS 211, superficial defects, including minor cracking and scaling, shall be repaired:

(a) using materials from the Recognized Product List;

(b) in accordance with the repair product manufacturer's recommendations; and

(c) to the satisfaction of the Ministry Representative.

582.32 Catch Basins – Catch basins shall be constructed, where shown and as required in the Drawings or as the Ministry Representative may direct, in accordance with the intent of SS Drawings SP582-02.01 through SP582-02.05.

582.33 Manholes – Manholes shall be constructed at the locations and to invert levels shown in Drawings or as the Ministry Representative may direct in accordance with the intent of SS Drawings SP582-03.01 through SP582-03.08.

582.34 Storm Drains – Storm drains shall be constructed where shown on Drawings, or as directed by the Ministry Representative, and in accordance with the Drawings and specifications covering the various types and as attached herewith. The trench and other preparatory work shall be approved by the Ministry Representative before actual placing starts.

582.35 582.35 Placing

582.35.01 Spigot and Bell Pipe – Spigot and bell pipe shall be laid beginning at the lower end with the bell end pointing up-grade. Suitable excavation or bedding must be provided to accommodate the bell so that the pipe is supported along its full length and not at the ends alone.

582.35.02 Concrete Pipe – Concrete pipe shall be laid beginning at the lower end with the groove end pointing up-grade. Pipe with elliptical reinforcement shall be laid with the minor axis of the reinforcement as marked by the manufacturer in a vertical position.

Concrete pipe joints shall be sealed using a rubber gasket, installed in accordance with the pipe manufacturer's recommendations.

582.35.03 Corrugated Steel Pipe – Corrugated steel pipe shall be laid beginning at the lower end and, where applicable, with the outside laps pointing up-grade and the longitudinal joint on the side. The separate sections shall be firmly jointed together and any metal in joints which is not thoroughly protected by galvanizing shall be coated with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

582.35.04 General – Pipes shall be accurately set and laid to even gradients, concentric and in straight lines between manholes.

582.36 Bedding and Backfilling – The bedding of all concrete pipes shall consist of 19 mm or 25 mm crushed gravel to a minimum depth of 150 mm below the pipe. The bedding below the middle 1/3 of the pipe shall be left uncompacted for a depth 75 mm or 1/24th of the pipe diameter, whichever is greater.

The bedding for corrugated steel pipe shall be in accordance with SS 303.

The backfill shall consist of 19 mm or 25 mm crushed gravel with a minimum cover of 150 mm and in the remainder of the trench shall consist of the natural excavated material unless such material is considered unsuitable by the Ministry Representative. Such natural material shall have a maximum size of 75 mm. All materials shall be compacted in horizontal layers not more than 200 mm.

582.37 Concrete End Walls – See SS 303.23.

582.38 Detectable Warning Mats – Detectable warning mats shall be constructed where shown on Drawings, or as directed by the Ministry Representative, and in strict accordance with the recommendations of the manufacturer.

MEASUREMENT

582.81 Portland Cement Concrete Curb and Gutter – Portland Cement Concrete Curb and Gutter will be measured by the METRE.

582.82 Catch Basins – Catch Basins will be measured per EACH.

582.83 Manholes – Manholes will be measured per EACH.

The Contractor shall allow in the bid price for the construction of manholes having an overall depth of concrete section of 2 m (see dimensions marked "variable"). In addition, the Contractor shall bid a price per 0.3 m of depth of manhole, which shall be added to or subtracted from the bid price for the manhole depending on the actual depth of the manhole to be constructed.

582.84 Storm Drains – Storm drains, including box culverts, will be measured by the LINEAR METRE in place.

582.85 Bedding and Backfill – Bedding and backfill will be measured by the CUBIC METRE in place.

582.86 Concrete End Walls – See SS 303.81.

582.87 Detectable Warning Mats – Detectable warning mats will be measured by the SQUARE METRE in place.

PAYMENT

582.91 Portland Cement Concrete Curb and Gutter – Payment for combined Portland cement CONCRETE CURB AND GUTTER will be at the Contract Unit Price per metre. The Contract Unit Price shall include everything furnished and done in connection with form setting, supply and placing of air entrained concrete; supply and installation of expansion joints; finishing, curing, protection and superintendence.

582.92 Catch Basins – Payment for CATCH BASINS will be at the Contact Unit Price per each. The Contract Unit Price shall include everything furnished and done; excavation, backfilling and tamping around catch basins, formwork, concreting, setting in of pipe drains, setting and securing of cast iron catch basin frames and gratings, steel baffles, reinstatement of surface, cleaning out, protection and superintendence and, all other things necessary.

582.93 Manholes – Payment for MANHOLES will be at the Contract Unit Price per standard 2 m manhole with appropriate depth correction as described in SS 582.83. The Contract Unit Price shall include everything furnished and done; formwork, concreting, setting in of storm drains, construction of smooth invert sections, cement mortar settings, steel ladder rungs, brick construction, setting of manhole frames and covers, cleaning out, protection and superintendence, and all other things necessary.

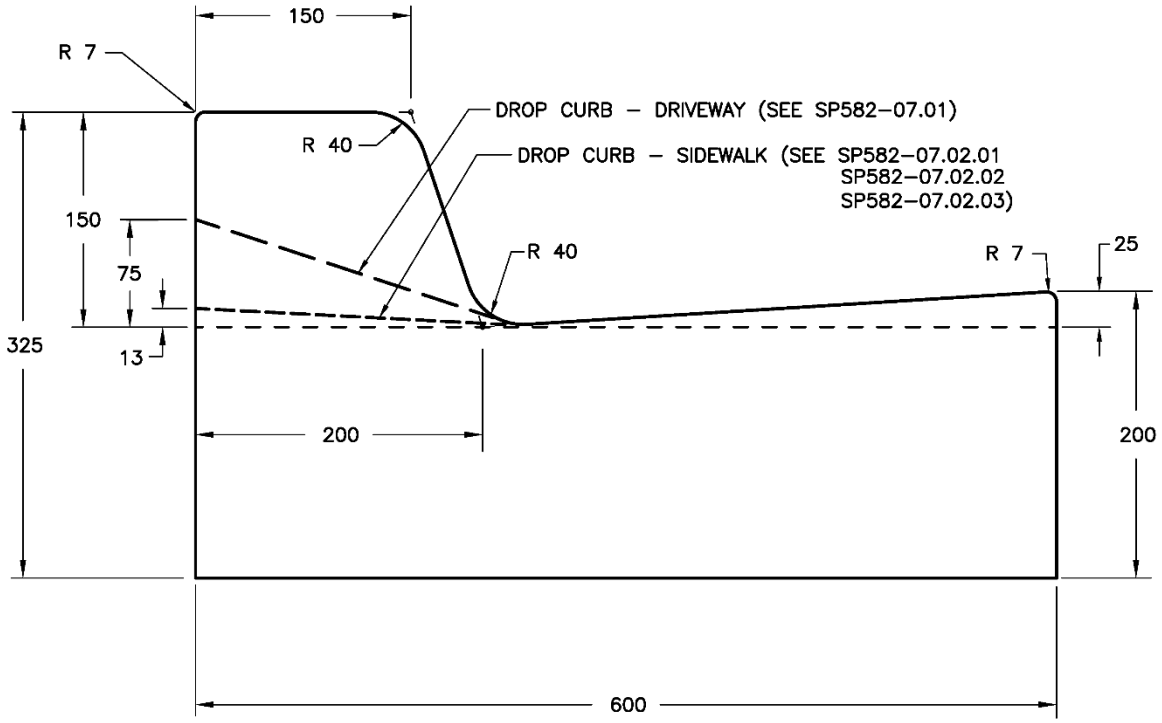
582.94 Storm Drains – Payment for STORM DRAINS will be at the Contract Unit Price per linear metre in place. The Contract Unit Price shall be accepted as full compensation for everything furnished and everything done in connection herewith, but shall not include payment for excavation, riprap, paving, concrete cradles, end walls, bedding and backfill. These shall be paid for at their respective prices in the Schedule of Approximate Quantities and Unit Prices.

582.95 Bedding and Backfill – Payment for BEDDING and BACKFILL will be at the Contract Unit Price per cubic metre in place.

582.96 Concrete End Walls – See SS 303.91.

582.97 Detectable Warning Mats – Payment for DETECTABLE WARNING MATS will be at the Contract Unit Price per square metre in place.

COMBINED CURB AND GUTTER	SP582–01.01

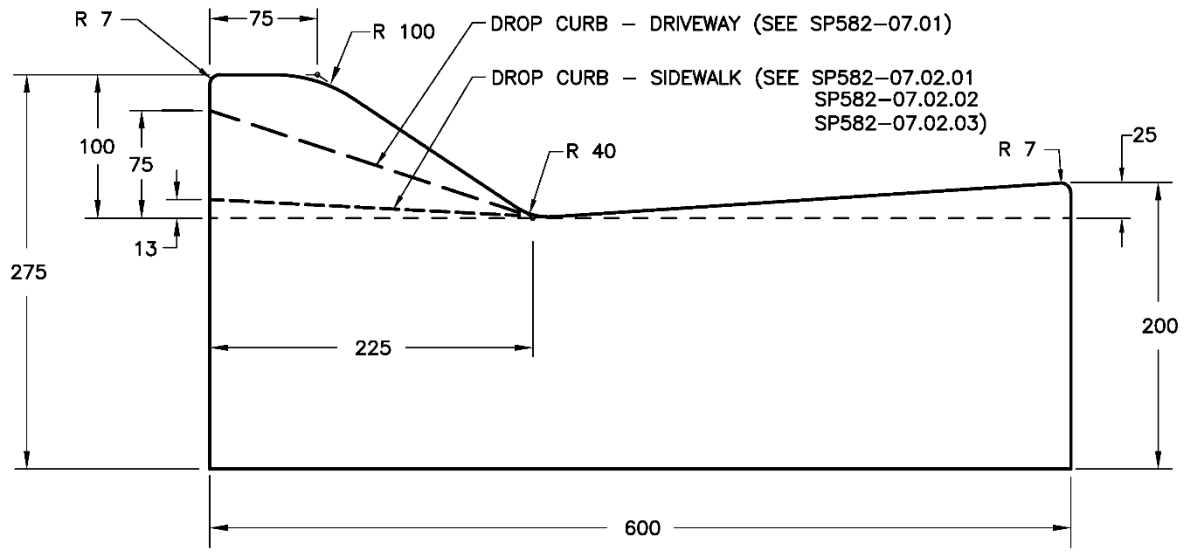


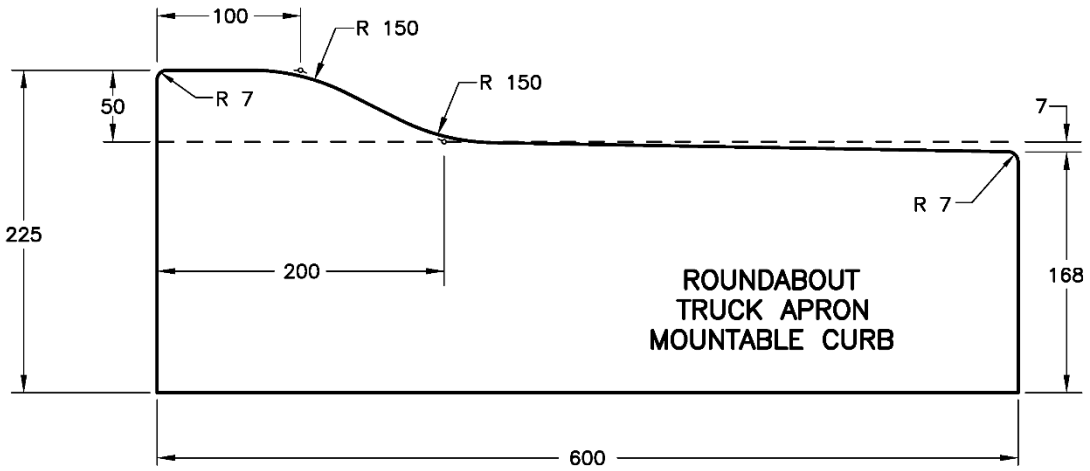
NOTE:
1. ALL CAST–IN–PLACE WORK TO CONFORM TO SS 582.17.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

MOUNTABLE CURB AND GUTTER	SP582–01.02



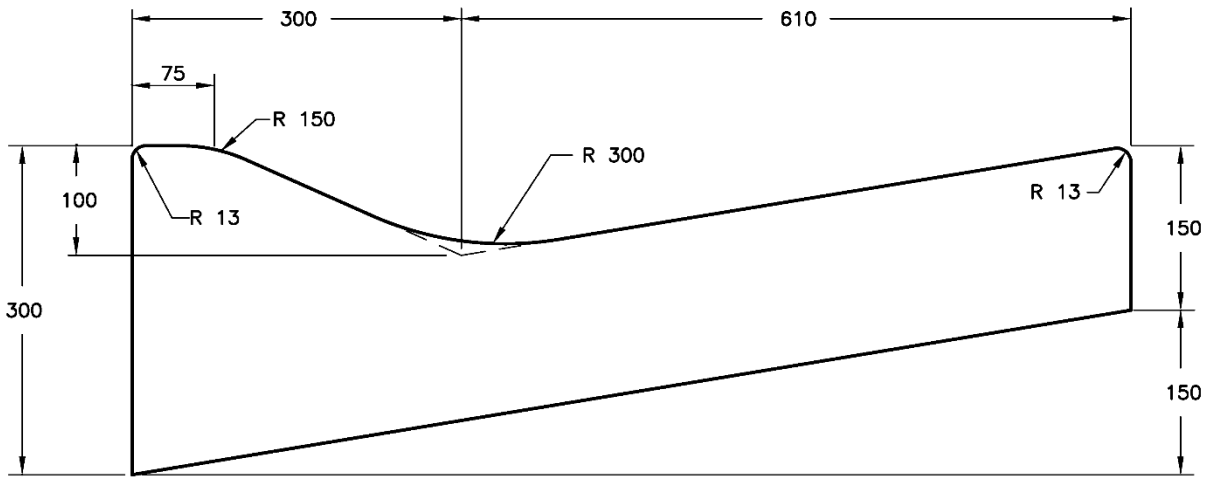


NOTE:
1. ALL CAST–IN–PLACE WORK TO CONFORM TO SS 582.17.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

VALLEY CURB AND GUTTER	SP582–01.03

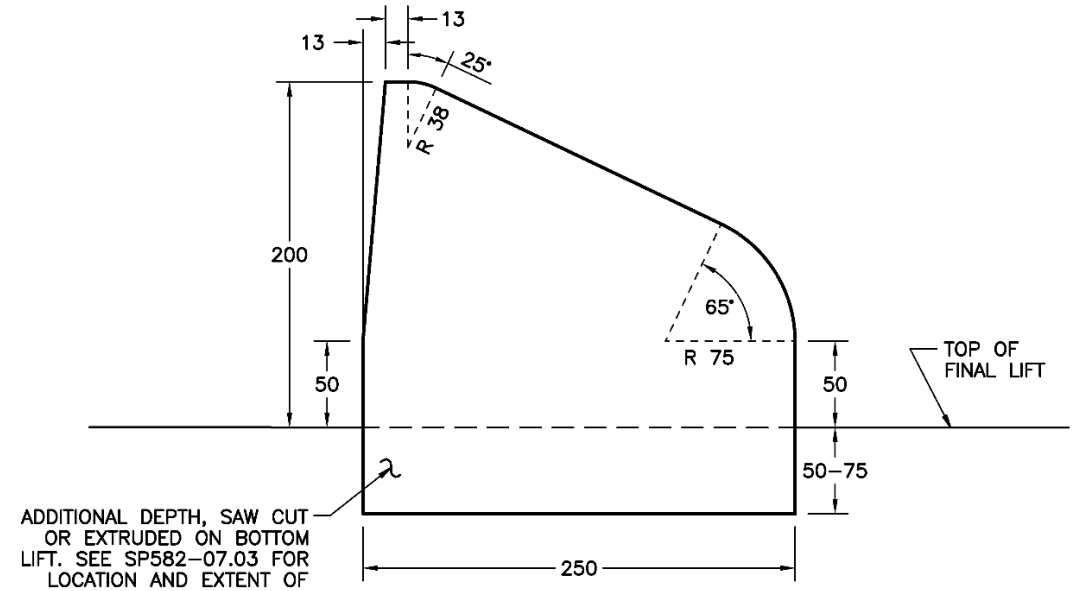


NOTE:
1. ALL CAST–IN–PLACE WORK TO CONFORM TO SS 582.17.

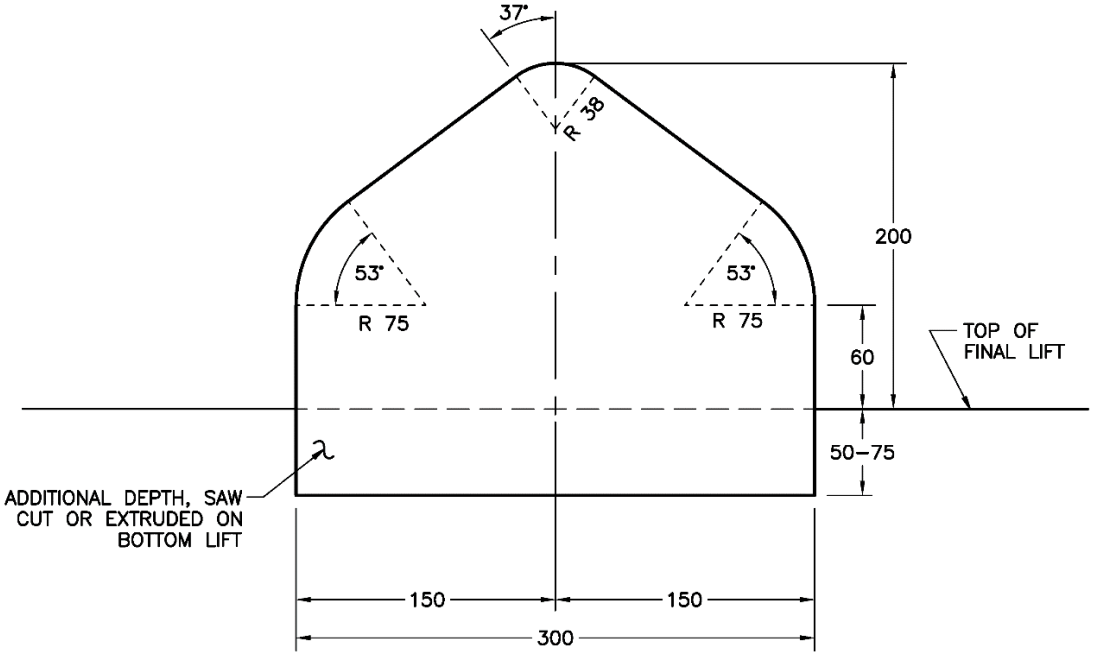
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

ISLAND OR MEDIAN CURB	SP582–01.04



EXTRUDED CONCRETE CURB
FOR ISLANDS & MEDIANS



EXTRUDED CONCRETE CURB
CENTER DIVIDER

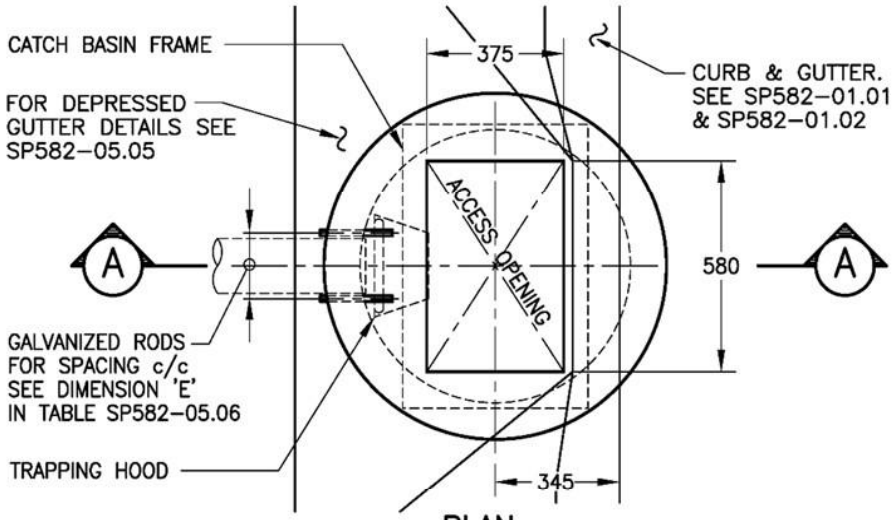
TYPICAL SECTIONS

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

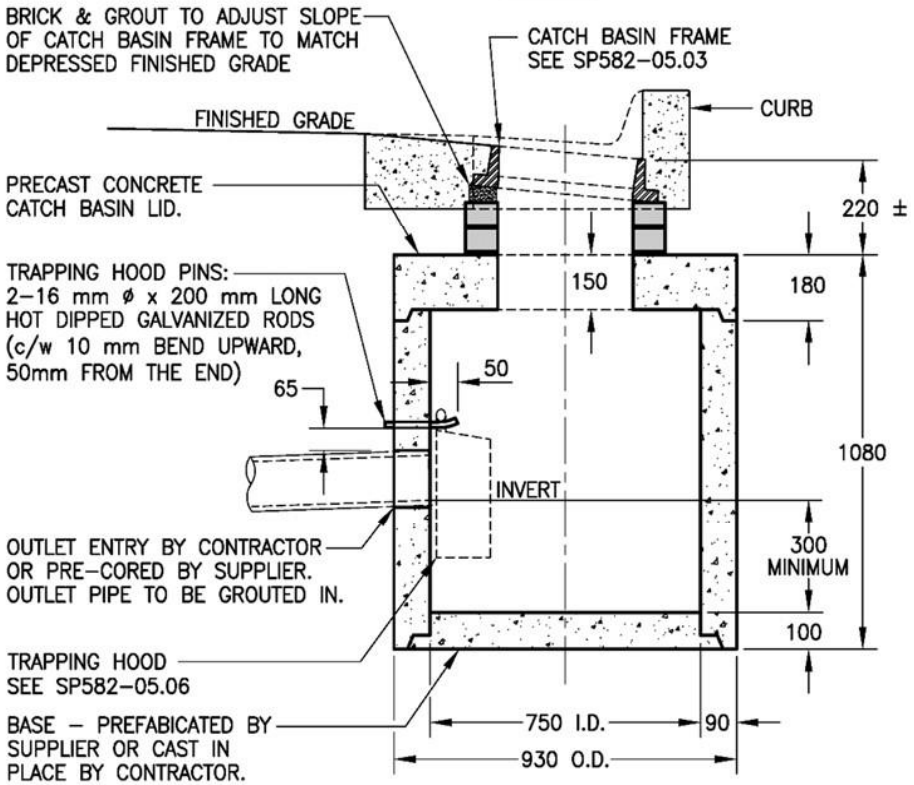
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

PRECAST REINFORCED CONCRETE CATCH BASIN

SP582–02.01



PLAN



SECTION A–A

NOTES:

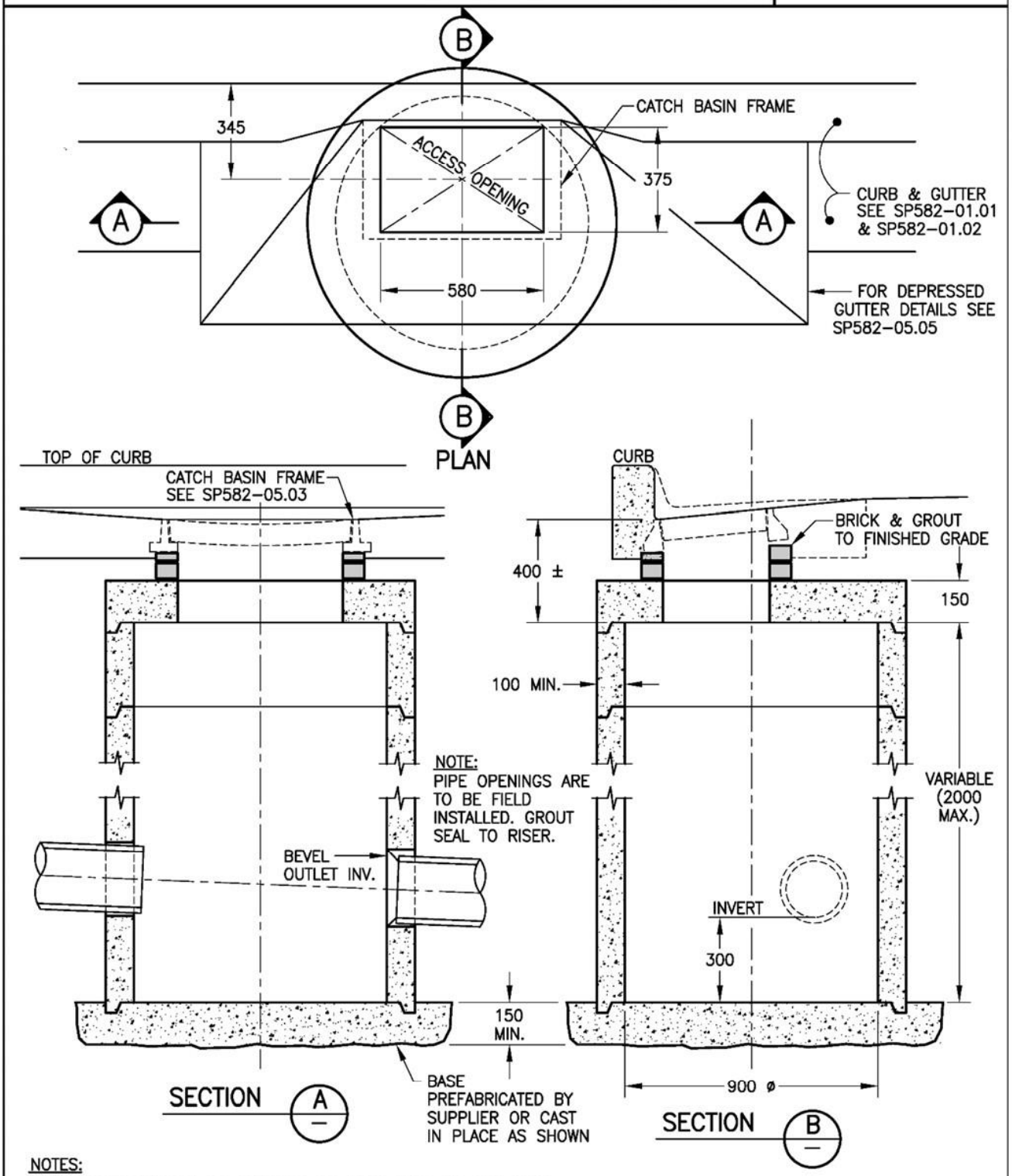
1. ALL PRECAST WORK TO CONFORM TO ASTM C478 OR CSA A257.
2. CATCH BASIN LID TO WITHSTAND HIGHWAY LIVE LOADING IN ACCORDANCE WITH THE APPLICABLE CLAUSES OF SS 582.
3. TRAPPING HOOD PINS SHALL BE SECURELY FASTENED TO THE PRECAST CONCRETE CATCH BASIN BARREL

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

PRECAST REINFORCED CONCRETE VARIABLE DEPTH CATCH BASIN

SP582–02.02



NOTES:

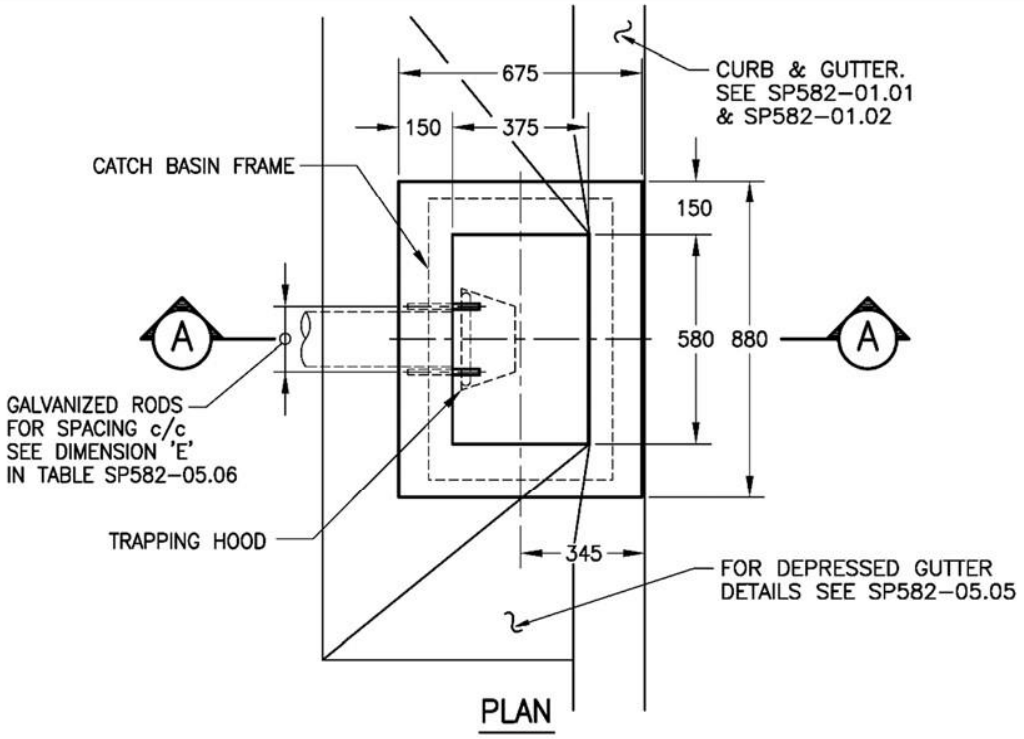
1. ALL PRECAST WORK TO CONFORM TO ASTM C478 OR CSA A257.
2. CATCH BASIN LID TO WITHSTAND HIGHWAY LIVE LOADING IN ACCORDANCE WITH THE APPLICABLE CLAUSES OF SS 582.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

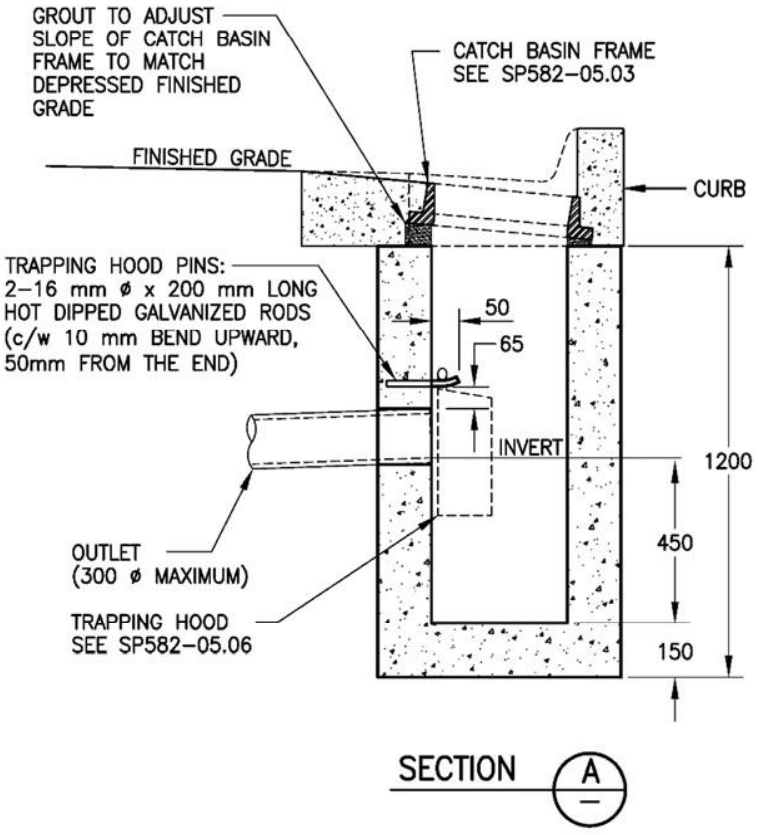
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

CAST IN PLACE CONCRETE CATCH BASIN

SP582–02.03



PLAN



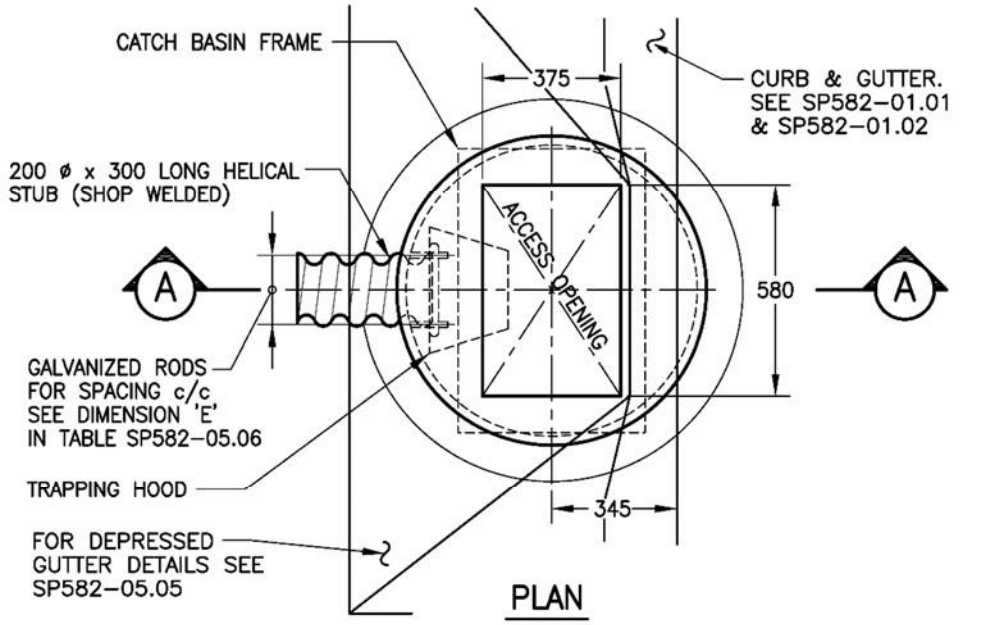
SECTION A–A

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

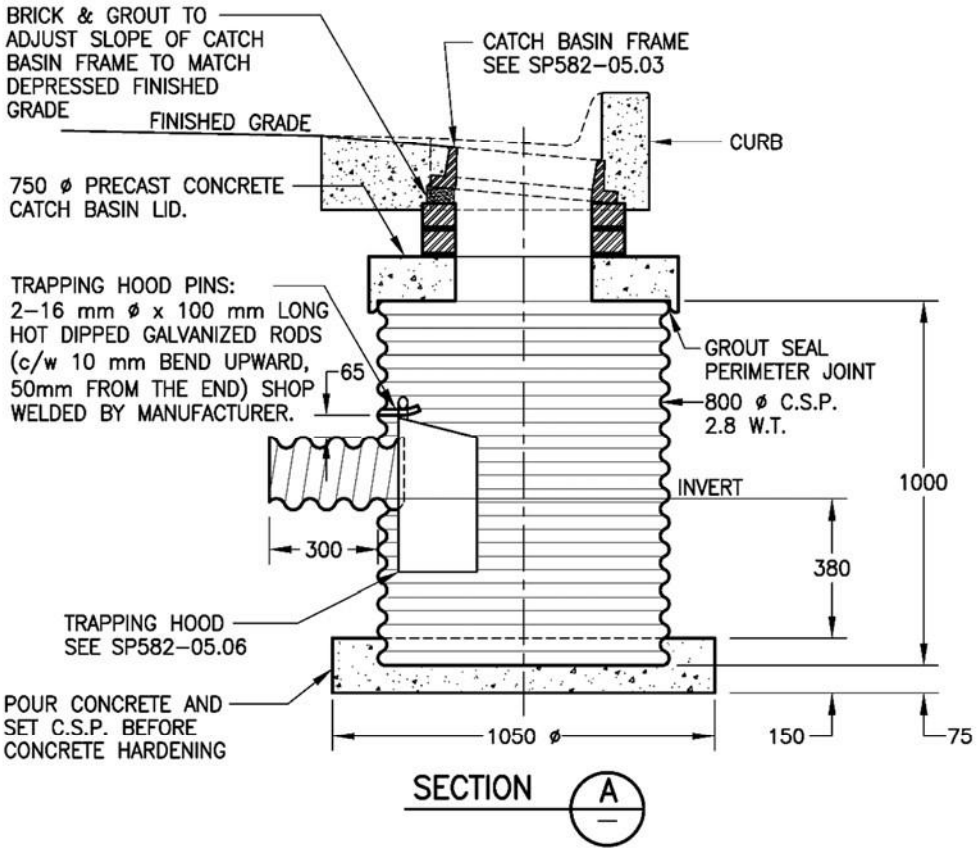
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

CORRUGATED STEEL CATCH BASIN

SP582–02.04



PLAN



SECTION A

NOTES:

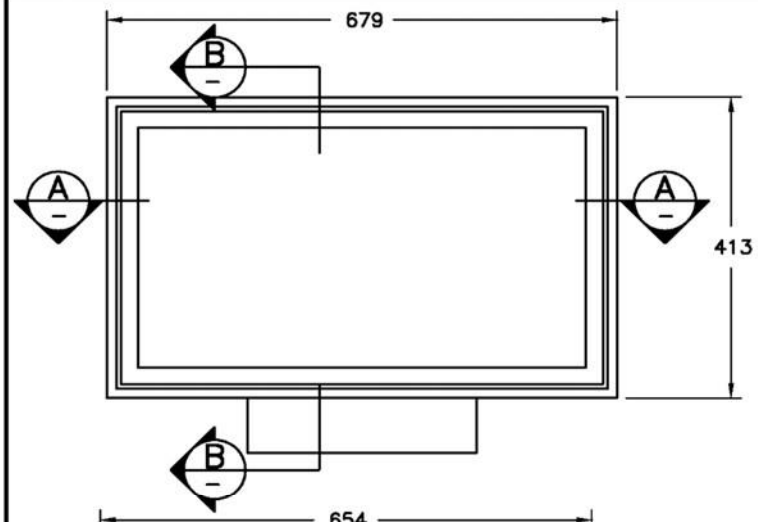
1. ALL PRECAST WORK TO CONFORM TO ASTM C478 OR CSA A257.
2. CATCH BASIN LID TO WITHSTAND HIGHWAY LIVE LOADING IN ACCORDANCE WITH THE APPLICABLE CLAUSES OF SS 582.

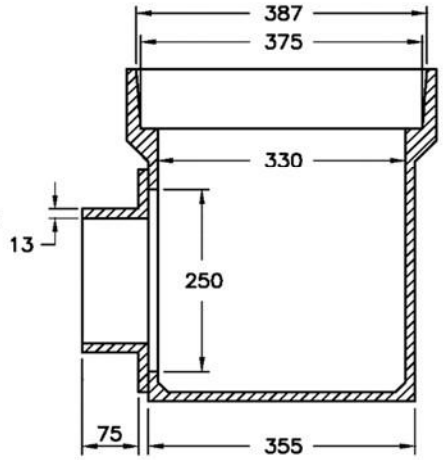
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

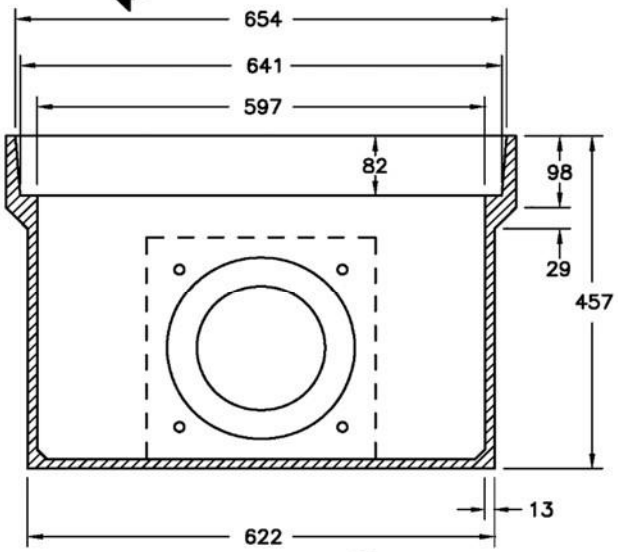
CAST IRON CATCH BASIN BOX AND ADAPTOR PLATE

SP582–02.05



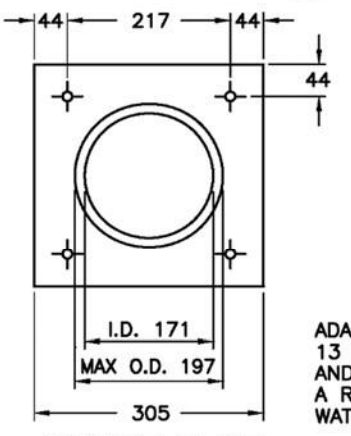


SECTION B



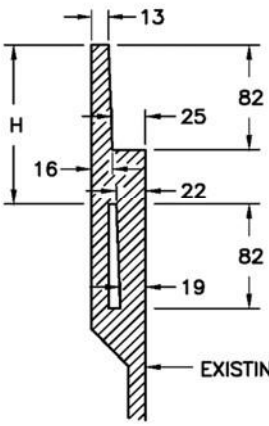
SECTION A

NOTE:
THIS CATCH BASIN BOX IS TO BE USED IN CONJUNCTION WITH A STANDARD CATCH BASIN GRATE (SEE SP582–05.01 OR SP582–05.02).
MATERIAL TO BE CAST STEEL OR CAST IRON AS ORDERED.
APPROXIMATE WEIGHT 136 kg



ADAPTOR PLATE

ADAPTOR PLATE TO BE DRILLED TO CLEAR 13 mm STUDS. THE FACE OF THE BOX AND PLATE TO BE GROUND SMOOTH, AND A RUBBER GASKET USED TO ENSURE A WATER TIGHT JOINT.



HEIGHT H TYPE A....100
TYPE B....125
TYPE C....150

OUTSIDE DIMENSIONS OF CAST IRON EXTENSION ARE 679 x 413

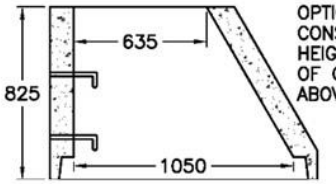
EXTENSION FOR CAST IRON CATCH BASIN BOX

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

PRECAST REINFORCED CONCRETE MANHOLE

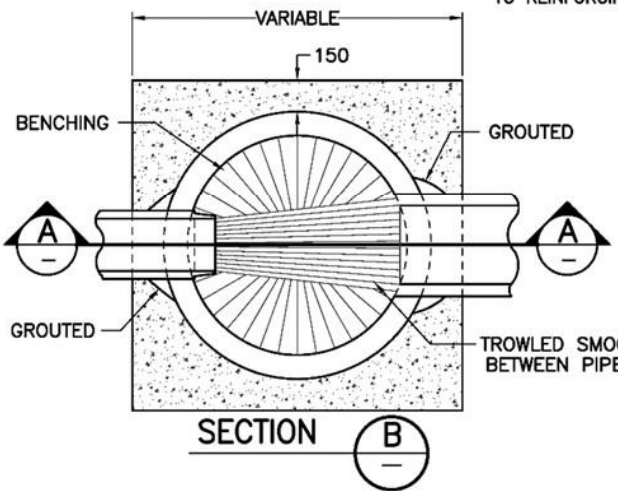
SP582–03.01



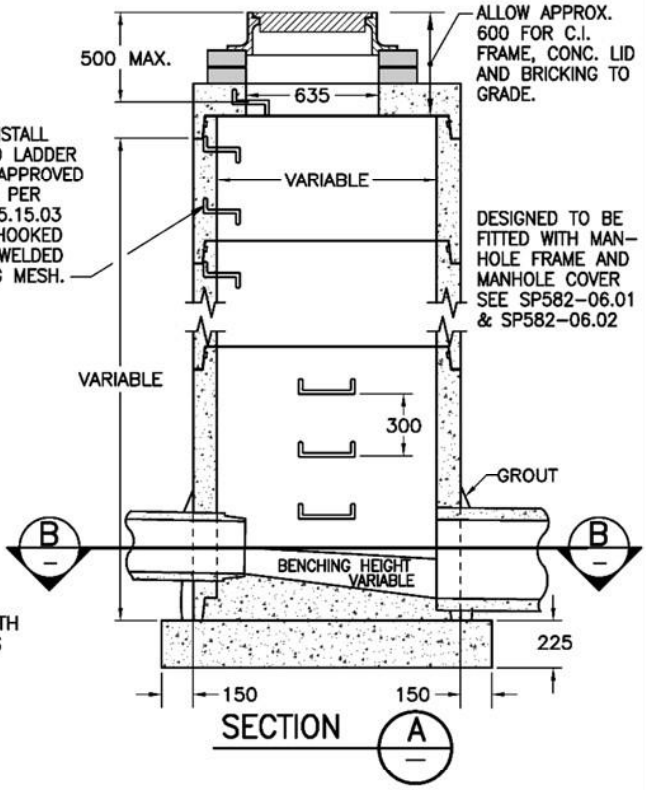
OPTIONAL BY DESIGNER. CONSIDERED ONLY WHEN HEIGHT ALLOWS BOTTOM OF CONE TO BE 2 m ABOVE INVERT.

1050 mm TO 625 mm MANHOLE REDUCING CONE

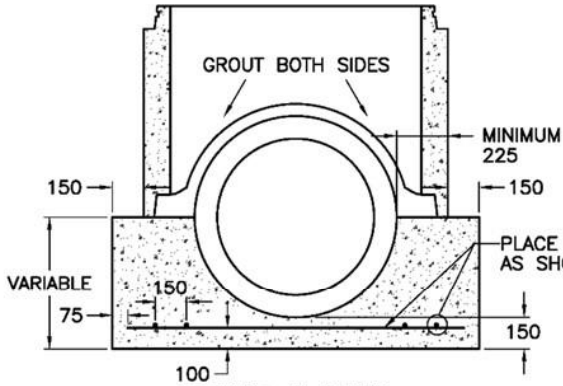
SUPPLIER TO INSTALL 19ø GALVANIZED LADDER RUNGS OR PREAPPROVED ALTERNATIVE AS PER SUBSECTION 145.15.03 RUNGS TO BE HOOKED AS SHOWN OR WELDED TO REINFORCING MESH.



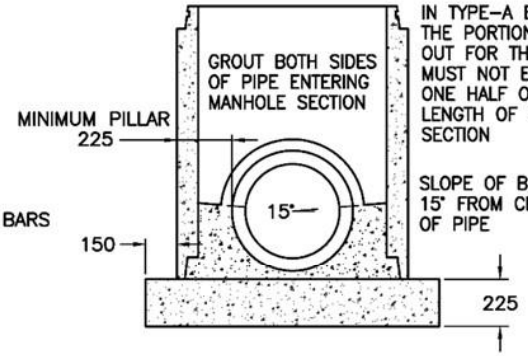
SECTION B



SECTION A



TYPE–B BASE
LARGE ø PIPES



IN TYPE–A BASE, THE PORTION BROKEN OUT FOR THE PIPES MUST NOT EXCEED ONE HALF OF THE LENGTH OF MANHOLE SECTION

SLOPE OF BENCH IS 15° FROM CENTRE OF PIPE

TYPE–A BASE
SMALL ø PIPES

CONSTRUCTION SEQUENCE

1. POUR A SQUARE CONCRETE BASE 225 mm THICK AND 300 mm WIDER THAN THE OUTSIDE OF THE MANHOLE SECTION.
2. BREAK PIPE ENTRANCES INTO MANHOLE SECTION, SET MANHOLE SECTION OVER THE PIPES ON THE CONCRETE BASE.
3. POUR A CONCRETE BENCH TO THE CENTRE POINT OF THE PIPE AND SLOPE OF 15°.
4. THROUGHLY GROUT AROUND PIPES ENTERING MANHOLES.

MANHOLE	MAX. PIPE	
INSIDE DIA.	BASE "A"	BASE "B"
750	375	375
900	530	530
1050	610	685
1200		760
1350		915
1500		1065
1800		1370

NOTES:

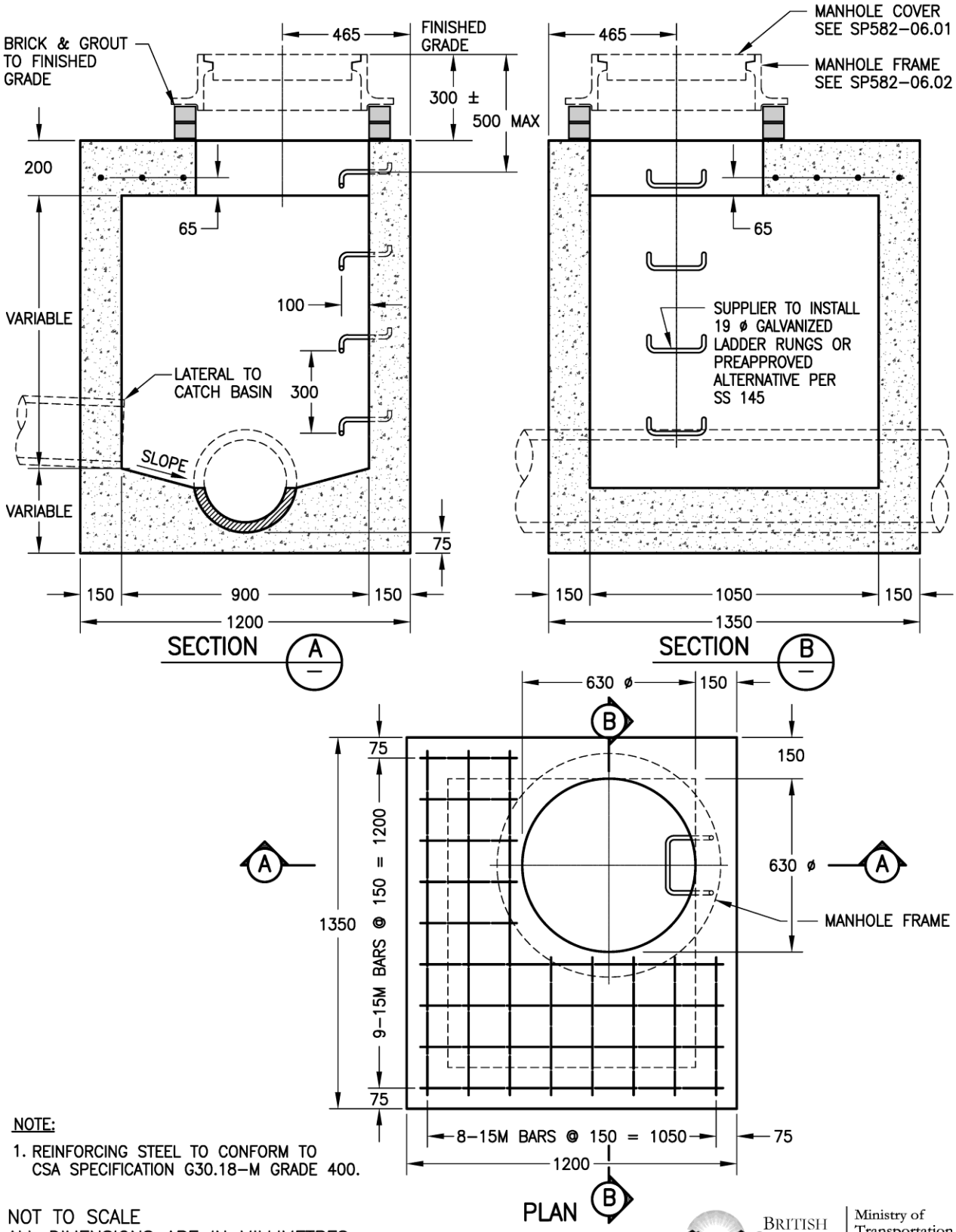
1. ALL PRECAST WORK TO CONFORM TO ASTM C478 OR CSA A257.
2. MANHOLE LID TO WITHSTAND HIGHWAY LIVE LOADING IN ACCORDANCE WITH THE APPLICABLE CLAUSES OF SS 582.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

CAST IN PLACE CONCRETE MANHOLE

SP582–03.02



NOTE:

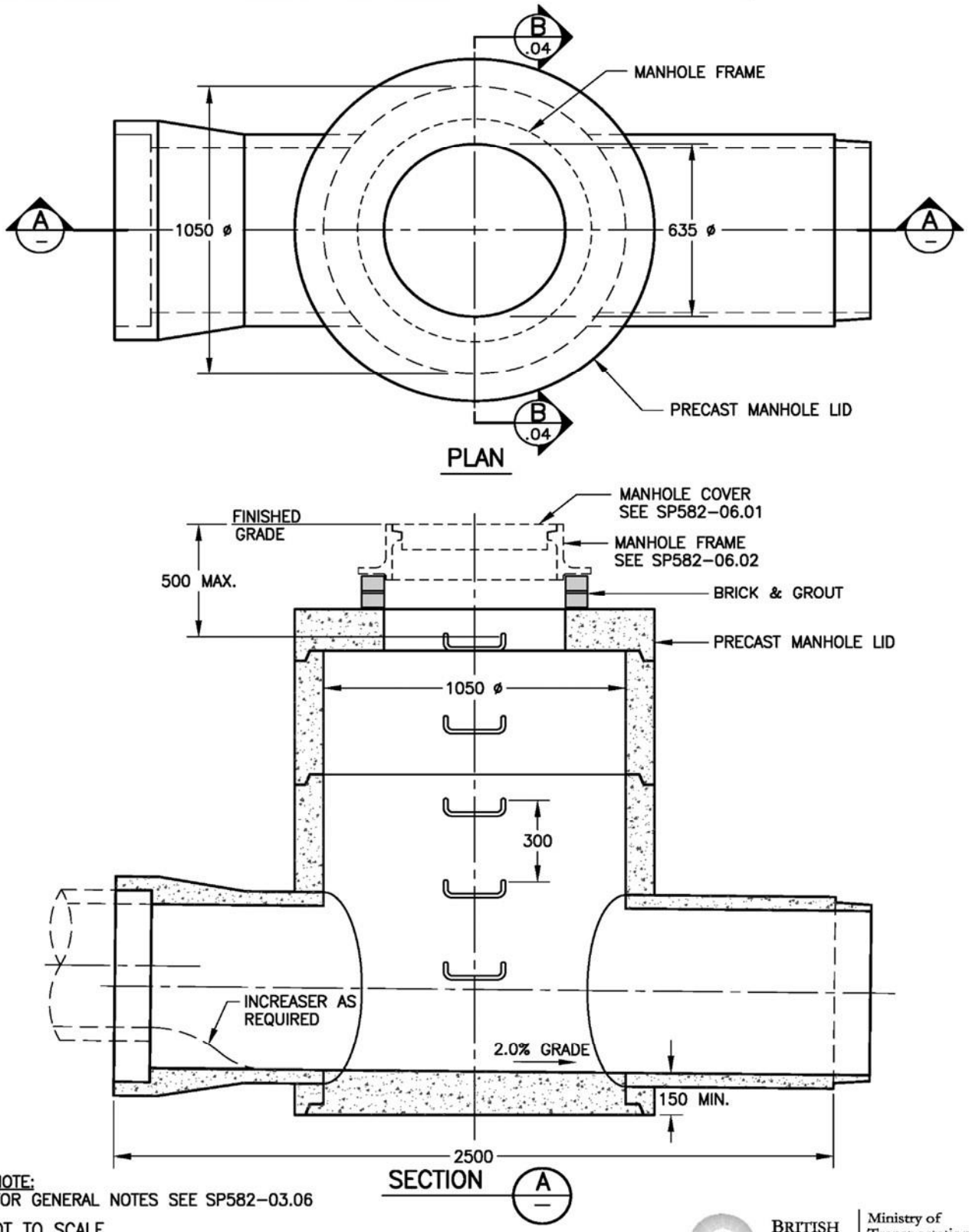
1. REINFORCING STEEL TO CONFORM TO CSA SPECIFICATION G30.18–M GRADE 400.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

PREFABRICATED CONCRETE TEE MANHOLES PIPE SIZES – 300 mm TO 675 mm

SP582–03.03



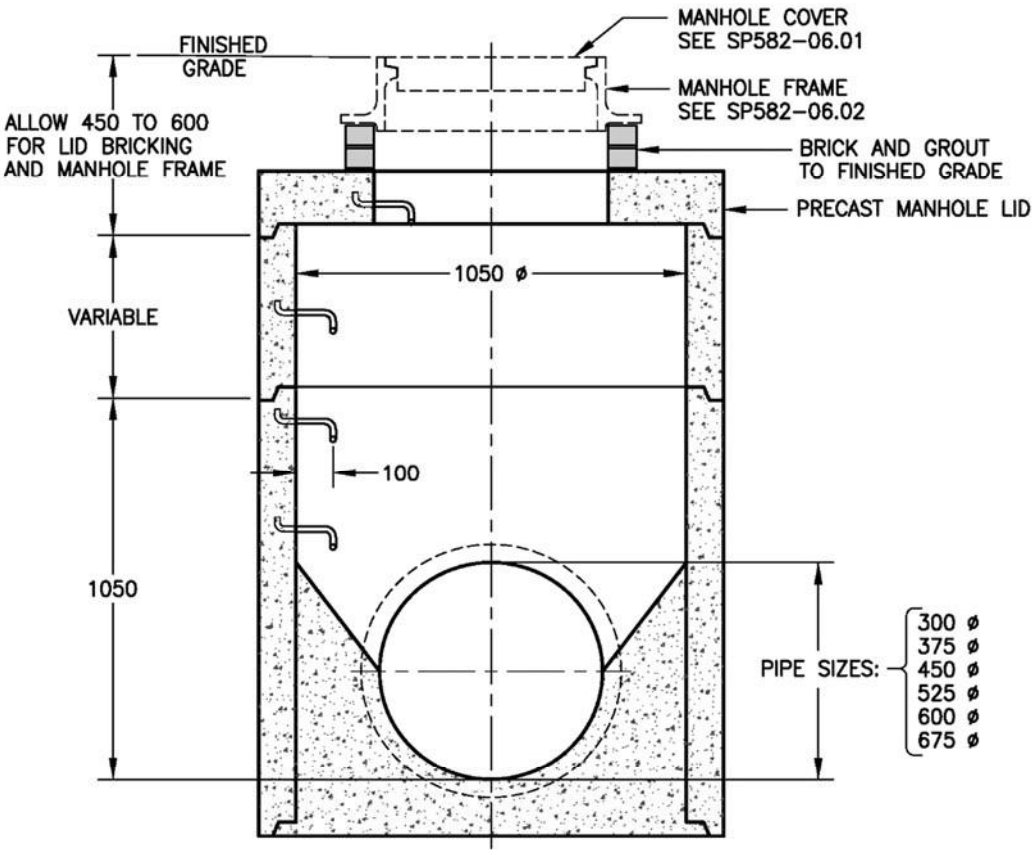
NOTE:
FOR GENERAL NOTES SEE SP582–03.06

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

PREFABRICATED CONCRETE TEE MANHOLES PIPE SIZES – 300 mm TO 675 mm

SP582–03.04



SECTION B .03

NOTE: FOR LOCATION OF SECTION SEE SP582–03.03

NOTE:
FOR GENERAL NOTES SEE SP582–03.06

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure